

3 13 14 : COS 214 Project Report

1. Plant Lifecycle and Nursery Operations

1.1 Commercial Nursery Workflow

Commercial nurseries operate through distinct phases that mirror natural plant development:

- **Planting/Propagation Phase:** Seeds or cuttings are initially planted in controlled environments
- **Nursery Phase:** Young plants receive intensive care to establish strong root systems and initial growth
- **Growing Phase:** Plants mature and develop their characteristic features
- **Sales Readiness:** Mature plants are prepared for retail sale
- **Withering/Decline:** Plants that receive inadequate care or exceed their optimal sale window deteriorate

This lifecycle model directly influenced our State Pattern implementation, where each plant transitions through distinct states (PlantedState, InNurseryState, GrowingState, ReadyForSaleState, WitheringState).

2. Plant Care Requirements

2.1 Watering Strategies

Different plant categories require distinct watering approaches:

- **Flood Irrigation:** Traditional method delivering large water volumes, suitable for trees and robust plants
- **Drip Irrigation:** Precise, water-efficient delivery for succulents and sensitive species
- **Misting:** Gentle moisture application for delicate flowers requiring humidity control
- **Standard Watering:** Regular watering for general nursery stock

This research informed our Strategy Pattern implementation with distinct watering strategies: WateringStrategy, FloodWateringStrategy, DripWateringStrategy, and GentleMistStrategy.

2.2 Fertilization

Commercial nurseries use controlled-release fertilizers with application amounts varying by plant type:

- Trees: Higher nitrogen requirements (80ml baseline)
- Flowers: Moderate feeding (50ml baseline)
- Succulents: Minimal fertilization (25ml baseline)

Implemented through FertilizingStrategy with species-specific nutrient amounts.

2.3 Pruning Methods

Horticultural pruning serves functional and aesthetic purposes:

- Minimal Pruning: Removal of dead/diseased material only (succulents)
- Standard Pruning: Shaping and deadheading (most flowers)
- Artistic Pruning: Deliberate sculpting for ornamental value (bonsai, topiary trees)

Implemented as MinimalPruningStrategy, StandardPruningStrategy, and ArtisticPruningStrategy.

3. Plant Species Categorization

3.1 Implemented Categories

- Flowers (e.g., Rose, Tulip, Orchid, Lily):
 - Water requirements: 150-250ml
 - Sunlight: Partial sun to full sun
 - Soil type: Well-drained, loamy
 - Lifecycle duration: approximately 20 seconds per state (simulated: 2-3 weeks real-time)
 - Care frequency: Water every 6s, fertilize every 15s, prune every 18s
- Trees (e.g., Bonsai, Oak, Maple, Pine):
 - Water requirements: 180-350ml
 - Sunlight: Full sun
 - Soil type: Well-drained, varies by species
 - Lifecycle duration: approximately 30 seconds per state (simulated: 4-6 weeks real-time)
 - Care frequency: Water every 10s, fertilize every 18s, prune every 16s

- Succulents (e.g., Aloe, Cactus, Echeveria, Jade):
 - Water requirements: 100-120ml
 - Sunlight: Bright indirect light
 - Soil type: Sandy, fast-draining
 - Lifecycle duration: approximately 25 seconds per state (simulated: 3-4 weeks real-time)
 - Care frequency: Water every 12s, fertilize every 20s, prune every 15s

These specifications were implemented through the FlowerProfile, TreeProfile, and SucculentProfile classes extending PlantSpeciesProfile.

4. Container Selection and Customization

4.1 Pot Materials

Nurseries offer various container materials, each with distinct properties:

- Clay/Terracotta: Porous, allows air circulation, classic aesthetic
- Plastic: Lightweight, moisture-retentive, economical
- Metal: Durable, modern aesthetic, heat-retentive
- Glass: Decorative, moisture-visible, fragile
- Wooden: Natural aesthetic, biodegradable options

Implemented through the Factory Method Pattern with ClayPotFactory, MetalPotFactory, GlassPotFactory, etc.

4.2 Customization Features

Retail nurseries offer pot customization for premium sales:

- Color finishes
- Decorative patterns
- Texture variations (glossy, matte, rustic)
- Additional features (drainage trays, decorative stones)

Implemented using the Decorator Pattern through ColorDecorator, PatternDecorator, TextureDecorator, and FeatureDecorator.

5. Inventory and Sales Management

5.1 Inventory Tracking

Modern nurseries maintain separate inventory systems:

- Greenhouse inventory: Plants in active cultivation
- Sales floor inventory: Plants ready for customer purchase
- Sold inventory: Transaction history and analytics

Implemented through the Singleton Pattern in InventoryManager with separate tracking vectors.

5.2 Staff Organization

Nurseries employ specialized teams:

- Greenhouse staff: Gardeners handling plant care (watering, fertilizing, pruning)
- Sales staff: Cashiers managing customer transactions and floor operations

Implemented using the Chain of Responsibility Pattern with distinct Gardener and Cashier handlers managed by StaffManager.

6. Design Decisions and Assumptions

6.1 Lifecycle Time Compression

Decision: Compressed plant lifecycle from weeks/months to seconds for demonstration purposes.

Rationale: Allows complete lifecycle observation within reasonable simulation timeframes while maintaining proportional state durations and care intervals.

Assumption: Time ratios between states and care requirements remain realistic (e.g., trees grow slower than flowers proportionally).

6.2 Withering as Terminal State

Decision: Plants that become neglected enter an irreversible WitheringState.

Rationale: Reflects commercial reality where severely neglected nursery stock becomes unsaleable and must be discarded.

Assumption: Recovery mechanisms (intensive care, rehabilitation) are beyond project scope.

6.3 Automated Care System

Decision: Commands are automatically generated and queued through the Observer pattern when plants need care.

Rationale: Simulates professional nursery operations where care schedules are systematic rather than reactive.

Assumption: Sufficient staff capacity exists; commands may queue but will eventually be processed.

6.4 Species-Specific Care

Decision: Each plant species has unique care requirements defined in its profile.

Rationale: Reflects botanical reality where species have evolved distinct environmental preferences.

Assumption: Care requirements within species are uniform (no individual plant variation implemented).

6.5 Container Customization Economics

Decision: Decorators add incremental costs to base pot prices.

Rationale: Reflects retail reality where customization commands premium pricing.

Assumption: Decoration costs are additive and independent (no bulk discount for multiple decorations).

7. Key Definitions

Lifecycle State: Distinct developmental phase with specific duration and care requirements

Care Strategy: Algorithmic approach to applying specific care type (water, fertilize, prune)

Species Profile: Configuration template defining care requirements and lifecycle parameters for a plant category

Greenhouse Inventory: Collection of plants under active cultivation, not yet available for sale

Sales Floor Inventory: Mature plants available for immediate customer purchase

Pot Decoration: Additional customization layer applied to base container (color, pattern, texture, features)

Care Interval: Time period between required care applications to prevent plant decline

8. References

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